

When More Motion Modeling Hurts: Causal Latency Compensation for Streaming Object Detection in Automated Driving

Antonio Pereira Jr.

Abstract—A detector output computed from an image at time t is applied at $t + \Delta$, after both the camera and scene objects have moved. We compared ego-motion-only propagation (B3) with B3 plus an uncertainty-damped causal object-motion estimate (B5) on a frozen Argoverse 2 cohort of 80 development, 20 model-selection, and 50 held-out logs, using prespecified parameters and a 300-ms high-yaw primary endpoint. B5 increased normalized center error relative to B3 by 19.18% for YOLO11n, 21.06% for YOLO11s, and 16.60% for RT-DETR-L; paired bootstrap intervals excluded zero. B3 also outperformed stale boxes, constant image velocity, SORT, ByteTrack, and OC-SORT predictions. Full-frame AP₇₅, recall, and false positives supported the localization result, whereas paired AP_{50:95} differences were inconclusive. Oracle future displacement improved on B3, showing that object motion is useful in principle, but causal velocity-sign and association errors dominated the deployed estimate. The B5 disadvantage grew from 2.58% at 100 ms to 30.40% at 500 ms, held for vehicles and for pedestrians and cyclists, and reproduced on a prospective extension cohort for all three detectors. A frozen causal gate never selected B5 in eligible extension comparisons. Measured ego-motion propagation is therefore the safer default for this system; object-motion augmentation should be enabled only by a reliability criterion validated on untouched data.

Index Terms—autonomous driving, streaming perception, object detection, ego-motion compensation, multi-object tracking, latency compensation, Argoverse 2, RT-DETR.

I. INTRODUCTION

OBJECT detectors are usually evaluated as if the scene stood still during inference. However, in an embodied system, a result computed from an image acquired at time t becomes actionable at $t + \Delta$, after both the camera and scene objects have moved. Streaming-perception benchmarks have formalized this mismatch and shown that offline accuracy overstates online performance [1]–[3]. This leaves a practical post-processing problem: transporting a delayed detector box into the camera frame that exists when the result arrives.

Image-space trackers extrapolate detections with constant-velocity or Kalman models [4], [5]; ByteTrack and OC-SORT add broader association and observation-centric updates [6], [7]. On an ego-moving platform, camera motion contributes an additional apparent displacement component, which trackers such as BoT-SORT and UCMCTrack approximate through image registration or a fixed ground-plane homography [8], [9]. Ego-motion-aware prediction methods therefore combine camera motion and depth with object-motion models to improve target-state extrapolation [10]. The expected hierarchy is clear:

correcting both ego and object motion should beat correcting ego motion alone.

An initial feasibility analysis on a small nuScenes subset suggested that object-motion augmentation could improve latency compensation when privileged instance-linked histories were available. We subsequently froze the modeling choices and evaluated both methods under common, deployable information constraints on a larger Argoverse 2 (AV2) cohort. Under this evaluation, the apparent benefit reversed: uncertainty-damped object-motion augmentation increased localization error relative to ego-motion-only propagation. Here we ask whether this reversal survives a transformer-based end-to-end detector, whether ego-only propagation is competitive with established causal tracker predictions, and whether full-frame current-time metrics and diagnostic analyses support the matched-object result.

This paper makes five contributions:

- a prespecified B3-versus-B5 reversal replicated across YOLO11n, YOLO11s, and the architecturally distinct RT-DETR-L;
- a comparison of both geometric models against stale, constant-velocity, SORT, ByteTrack, and OC-SORT baselines under identical delayed-detection conditions;
- separate persistent-object and full-frame target-time evaluations that bound what box propagation can and cannot fix;
- failure localization through oracle component substitutions and fixed diagnostic categories; and
- a one-shot evaluation of a previously frozen causal gate on untouched extension data.

Oracle displacement confirmed that object motion contains useful corrective information, while universal application of its noisy causal estimate increased error.

II. RELATED WORK

A. Latency-aware perception

Streaming accuracy evaluates a perception stack at the time its output becomes available rather than at image acquisition [1]. Later work trained detectors to forecast future-frame detections, added long–short temporal fusion and distillation, and benchmarked complete vision-centric streaming systems [2], [3], [11], [12]. These forecasting detectors require retraining and absorb latency inside the network. Our analysis targets a narrower deployment layer: transporting fixed detector boxes through a known latency without retraining the detector.

B. Tracking-by-detection and motion prediction

SORT uses a constant-velocity Kalman filter with Hungarian assignment [5], [13]. ByteTrack associates both high- and

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low-confidence detections to preserve trajectories [6]. OC-SORT applies observation-centric velocity updates to limit error accumulation after missed observations [7]. BoT-SORT and UCMCTrack add camera-motion compensation estimated from image registration or a calibrated ground-plane homography [8], [9]; the B3 model studied here can be read as the measured-pose, lidar-depth limit of that compensation family. Ego-motion-aware target-prediction methods further attempt to separate object dynamics from apparent motion induced by the moving camera, using camera motion and depth to improve causal target extrapolation [10]. These methods provide stronger causal comparators than a stale box or a two-frame velocity estimate, and they expose the central uncertainty asymmetry of the problem: ego motion is shared across a frame and measured from platform poses, whereas object motion depends on object-specific association, depth, and detector stability over time.

C. Autonomous-driving datasets and detectors

AV2 provides synchronized cameras, lidar, ego poses, calibration, and tracked annotations [14]. nuScenes served only for secondary harmonization [15]. YOLO11n and YOLO11s supplied two detector capacities through the Ultralytics implementation [16]; RT-DETR-L supplied an end-to-end transformer detector without non-maximum suppression as an architecture-level robustness test [17].

III. PROBLEM FORMULATION

Let b_t be a detector box computed from an image acquired at time t and delivered after latency Δ . The target is the box geometry at $t + \Delta$. B3 back-projects the source box through source-time depth, transforms it with measured ego poses, and reprojects it into the target camera. B5 adds an uncertainty-damped estimate of object displacement obtained causally from associated detector history.

For detector d and log l , the primary error is the normalized center distance

$$e_{l,d} = \frac{\sqrt{(\hat{c}_x - c_x)^2 + (\hat{c}_y - c_y)^2}}{\sqrt{W^2 + H^2}}, \quad (1)$$

where (W, H) is image size. The primary estimand is the median across logs of the paired within-log relative difference

$$r_l = \frac{\tilde{e}_{l,B5} - \tilde{e}_{l,B3}}{\tilde{e}_{l,B3}}. \quad (2)$$

Positive values indicate harm from object-motion augmentation.

IV. METHODS

A. Frozen cohorts and detector outputs

Before evaluating the held-out logs, we fixed the cohort-selection seed, modeling parameters, analysis definitions, and the 300-ms high-yaw primary endpoint. Table I records these frozen choices.

The fixed AV2 cohort contained 80 development, 20 model-selection, and 50 held-out validation logs, selected before outcome analysis with seed 20260717 using pre-outcome city, ego-speed, yaw, and traffic-density metadata. Only the forward

ring camera, lidar, calibration, poses, and annotations were required.

All detectors ran at 640-pixel inference size and confidence threshold 0.25. RT-DETR-L used the `rt detr-l.pt` checkpoint distributed through the Ultralytics framework; SHA-256 hashes of all three detector checkpoints are recorded in the repository manifest. The RT-DETR-L held-out run comprised 50 logs and 16,022 frames. Detector outputs were checkpointed before propagation. COCO detections were mapped to vehicle, pedestrian/cyclist, and other groups.

A common threshold does not equalize output density across architectures. At the frozen endpoint RT-DETR-L emitted 20.5 detections per frame against 7.9 for YOLO11n and 9.8 for YOLO11s, with a lower median confidence (0.476 against 0.521 and 0.606) and 40.6% of detections below 0.4. Absolute false-positive counts therefore differ across detectors for reasons that precede propagation, which is why all method contrasts are paired within detector and log.

Latency horizons were 100, 200, 300, 400, and 500 ms. The frozen primary endpoint was 300 ms in the upper development-cohort quartile of absolute yaw rate (0.0170 rad/s). Forty held-out logs contained eligible matched comparisons at this endpoint for all three detectors; all 50 contributed to all-yaw analyses.

B. Causal propagation and tracker baselines

Seven methods were evaluated on identical eligible comparisons:

- T0: stale source box;
- T1: constant image-space velocity;
- T2: SORT Kalman prediction;
- T3: ByteTrack causal prediction;
- T4: OC-SORT causal prediction;
- T5/B3: measured ego-motion-only propagation;
- T6/B5: B3 plus uncertainty-damped estimated object motion.

Deployable methods saw only source-time detections and histories: no target-time detections, future annotation geometry, future identity, or future velocity entered their predictions.

B3 used source-box lidar depth, source and target ego poses, and calibrated camera projection. B5 estimated object velocity from same-class associated detector histories over at most 600 ms and four observations, multiplied object displacement by a frozen reliability weight based on lidar support, depth dispersion, and track age, and capped speed at 30 m/s. Table I lists all frozen modeling and analysis parameters; they were fixed with the method specification before any AV2 outcome was computed and were not retuned afterward.

C. Persistent-object and full-frame evaluation

Matched-object analysis evaluated objects visible at both source and target times, reporting normalized center error, IoU, and recall at IoU 0.5. Full-frame evaluation compared all propagated detections with all target-time annotations, reporting $AP_{50:95}$, AP_{50} , AP_{75} , precision, recall, false positives per frame, new-object miss fraction, and disappeared-object false-positive

TABLE I
FROZEN MODELING AND ANALYSIS PARAMETERS

Parameter	Value
Cohorts and endpoint	
Cohort selection seed	20260717
Development / model-sel. / held-out logs	80 / 20 / 50
Extension logs (gate; freeze seed 20260719)	45
Latency horizons	100–500 ms, step 100
Primary endpoint	300 ms, high yaw
High-yaw threshold ^a	$ \dot{\psi} \geq 0.0170$ rad/s
Detector inference	
Input resolution	640 px
Confidence threshold	0.25
Class mapping	vehicle, ped./cyclist, other
History association (B5)	
Assignment	Hungarian, same class only
Minimum IoU	0.10
Max. normalized center distance	0.15
Maximum track gap	600 ms
History length	4 observations
Object-motion estimate (B5)	
Motion model	const. velocity, 3-D history
α – β smoothing	$\alpha = 0.65$
Speed cap	30 m/s
Reliability damping (B5)	
Lidar-support saturation	10 points
Depth-dispersion scale	5.0 m
Track-age saturation	3 observations
Depth recovery	
Source	source-time lidar inside box
Statistic / uncertainty	median / scaled MAD
Minimum points; fallback	1; box-geometry regression
Statistics	
Independent unit	complete log
Bootstrap	10,000 paired log-cluster repl.
Matched-object recall threshold	IoU 0.5
Frozen causal gate	
Model	ℓ_2 logistic regr. ($C = 1.0$)
Features	13 causal source-time features
Decision threshold	0.60 (grid 0.50–0.90) ^b

^a Upper quartile of absolute yaw rate in the development cohort.

^b Selected on model-selection logs under a 2-percentage-point recall-loss constraint; frozen before extension outcomes.

fraction. The scopes are reported separately because matched-object localization does not measure objects entering or leaving the frame during latency.

D. Component substitutions and diagnostic mechanisms

After the primary result was frozen, post hoc substitutions replaced one deployed component at a time with evaluation-only oracle information: association, historical boxes available no later than source time, historical three-dimensional positions, source depth, and future object displacement. The last substitution tests whether object motion can help in principle; it is nondeployable and was excluded from method-success decisions.

For objects harmed by B5, a fixed hierarchy assigned one of: incorrect association, velocity-sign error, velocity-direction error, velocity-magnitude error, detector-box jitter interpreted as motion, unstable short-track extrapolation, depth-induced

projection error, missed acceleration, clipping/boundary error, distance-dependent failure, or unclassified. These labels are diagnostic and may reflect interacting causes. An independent post hoc direction-agreement analysis compared the estimated two-dimensional ground-plane velocity with the oracle source-to-target displacement. Objects with true speed below 0.1 m/s were treated as directionally ambiguous; among the remainder, a negative vector dot product indicated that the estimate pointed into the opposite half-plane. Oracle displacement was used only for diagnosis and was unavailable to deployable predictions.

E. Sensitivity analysis and frozen causal gate

Sensitivity analyses stratified results by latency, yaw, distance, detector confidence, track age, association confidence, velocity-sign consistency, class, and detector. Latency and detector strata were confirmatory; the rest were exploratory. Class-specific estimates used complete logs as the independent units and the same paired log-cluster bootstrap as the primary analysis.

The frozen object-level diagnostic table carries the instrumented estimator internals for YOLO11n and YOLO11s only. Class-stratified and image-space direction results for RT-DETR-L were therefore recomputed from the frozen low-level prediction and target tables, which record predicted geometry, target geometry, annotation category, and yaw rate for all three detectors. The reconstruction was validated by running it on the YOLO detectors and comparing against the frozen values: log counts, comparison counts, median errors, and paired relative differences agreed to a maximum absolute difference of 1.7×10^{-18} , and per-class comparison counts summed to the totals in Table II. Analyses that depend on estimator internals unavailable for RT-DETR-L, namely the hierarchical mechanism labels and the ground-plane velocity comparison, remain restricted to the YOLO pipelines.

A previously frozen causal gate was evaluated once on available extension outputs. The gate used only source-time or causal-history features, was audited for leakage, and selected between B3 and B5 without threshold retuning. Success required lower center error than B3, noninferior IoU, at most two percentage points of recall loss, improvement in at least 70% of extension logs, and a paired bootstrap interval excluding zero. RT-DETR prospective extension inference was incomplete and excluded.

F. Statistics, runtime, and audits

Logs were the independent units. Principal contrasts used paired absolute and relative differences, the percentage of logs improved, 10,000 paired log-cluster bootstrap replicates, and Wilcoxon signed-rank sensitivity tests. Full-frame AP was calculated both as pooled target-time detection performance and as paired log-level contrasts, with the aggregation level stated explicitly.

Audits checked detector identity, cohort overlap, timestamp order, complete T0–T6 model sets, duplicates, invalid coordinates, future-information flags, and independent headline recalculation. The final closure contained 100 valid current-detector scientific units plus four shard-summary rows, no duplicate or missing units, a provenance PASS, and a leakage

PASS WITH LIMITATIONS reflecting incomplete extension coverage rather than leakage in the core held-out analysis. Twenty analysis-closure tests passed.

V. RESULTS

A. The reversal generalized across detector architectures

At 300 ms in high-yaw driving, B5 increased normalized center error relative to B3 for all three detectors: 19.18% for YOLO11n, 21.06% for YOLO11s, and 16.60% for RT-DETR-L (Table II; Fig. 1). All paired bootstrap intervals excluded zero, and B5 improved only 15.0%, 20.0%, and 17.5% of eligible logs, respectively.

In the all-yaw confirmatory analysis, harm grew monotonically with horizon: 2.58% at 100 ms, 8.77% at 200 ms, 15.50% at 300 ms, 23.03% at 400 ms, and 30.40% at 500 ms—the signature of extrapolation error accumulating with time.

B. B3 outperformed strong causal tracker predictions

B3 had the lowest normalized center error among T0–T6 for every detector at the primary endpoint (Fig. 2). OC-SORT, the strongest alternative, remained 73.34% worse than B3 for YOLO11n, 72.01% for YOLO11s, and 56.23% for RT-DETR-L, with paired bootstrap intervals excluding zero. Constant image velocity, SORT, and ByteTrack trailed further. These baselines were evaluated as causal box-prediction modules under identical delayed-detection conditions, not as complete tracking systems, so the comparison does not rank full MOT pipelines.

C. Full-frame evaluation supported localization but $AP_{50:95}$ was mixed

Full-frame evaluation agreed with the matched-object result where the two scopes overlap (Fig. 3). At the frozen endpoint, B5 reduced AP_{75} by 20–28% across detectors, reduced recall, and increased false positives per frame; paired log-level AP_{75} intervals excluded zero for both YOLO detectors and were borderline for RT-DETR-L. Paired $AP_{50:95}$ intervals included zero for all three detectors, and pooled RT-DETR-L $AP_{50:95}$ was nearly identical between methods (0.04661 for B3, 0.04685 for B5), showing how pooled and paired log-level aggregation can diverge. Absolute AP levels are low because propagated boxes are scored against every target-time annotation, including the small and newly visible objects that no propagation method can produce, under a COCO-to-AV2 class mapping at 640-pixel inference. The estimand throughout is the paired within-log contrast between two propagation methods on identical detections, which these factors leave unaffected.

New-object miss fractions were high for both methods: objects entering the frame after the source time cannot be recovered by box propagation. This limitation is shared by all propagation-only methods and is why the two evaluation scopes are reported separately.

D. Oracle substitutions showed useful signal obscured by estimation error

Replacing deployed association with oracle identity reduced B5 error by 9.66% for YOLO11n and 11.42% for YOLO11s. Replacing detector-history boxes with true causal historical boxes removed a further 17.40% and 18.77%. Oracle future displacement improved on B3 by 11.11% and 9.85% (Fig. 4). Correct object displacement therefore helps in principle; the deployed estimator failed because its association and velocity errors exceeded the residual displacement it was correcting.

E. Failure mechanisms and exploratory operating strata

Velocity-sign error was the leading mechanism, covering 37.16% of harmed YOLO11n objects and 38.67% of harmed YOLO11s objects, followed by incorrect association at 24.44% and 22.79%. Velocity-magnitude and direction errors each contributed 14–16%; depth-induced projection errors were rare under the fixed hierarchy.

Harm was not uniform across operating conditions. For objects at 0–20 m, B5 reduced center error by 8.28% and improved 69% of logs; in the high-confidence detector stratum it reduced error by 15.76% and improved 65% of logs. Beyond 40 m it was 26.85% worse, and 34.58% worse at low confidence (Fig. 5). Selective compensation is therefore plausible, but these strata are exploratory and do not validate the current selector.

F. Class-specific effects and motion-direction agreement

The class-stratified analysis showed that the primary reversal was not an artifact of pooling road-user categories (Table III). Vehicles, which constituted most eligible comparisons, showed a consistent B5 disadvantage across all three detectors: 20.48% for YOLO11n (95% bootstrap interval 11.72–37.21%), 19.01% for YOLO11s (10.37–30.08%), and 15.92% for RT-DETR-L (7.68–22.47%).

Pedestrian and cyclist estimates were imprecise under the YOLO detectors, which recovered only 527 and 894 eligible comparisons: 2.60% for YOLO11n with an interval touching zero (0.00–16.80%) and 16.63% for YOLO11s over a wide interval (0.28–65.80%). RT-DETR-L resolved this stratum. Its higher detection density yielded 1872 pedestrian/cyclist comparisons across 29 logs, and the paired disadvantage was 26.64% with an interval excluding zero (1.49–44.53%). Vulnerable road users therefore follow the same direction as vehicles, and the earlier imprecision reflected detector recall on small objects rather than a class-specific exception. No class in any detector provided reproducible evidence that universal B5 application improved localization.

Two independent direction checks corroborated the hierarchical failure labels. In the ground-plane check, after excluding objects with true speed below 0.1 m/s, the estimated motion pointed into the same half-plane as the oracle displacement for 51.08% of YOLO11n objects and 51.41% of YOLO11s objects. Near-stationary objects were common and directionally ambiguous, accounting for 45.25% of YOLO11n and 43.65% of YOLO11s diagnostic objects. Even when displacement was large enough to classify direction, the causal estimate was near chance at selecting the correct half-plane.

TABLE II
FROZEN 300-MS HIGH-YAW HELD-OUT RESULT

	YOLO11n	YOLO11s	RT-DETR-L
Eligible logs	40	40	40
Object comparisons	11,425	14,089	25,729
B3 normalized center error	0.007003	0.005843	0.008512
B5 normalized center error	0.009146	0.007645	0.010176
Paired B5 disadvantage	19.18%	21.06%	16.60%
95% bootstrap interval	10.80–37.05%	8.30–31.51%	11.93–23.38%
B3 median IoU	0.5550	0.5480	0.3647
B5 median IoU	0.5270	0.5026	0.3070
Logs favoring B5	6/40	8/40	7/40
Wilcoxon p	0.0016	0.0006	0.0010

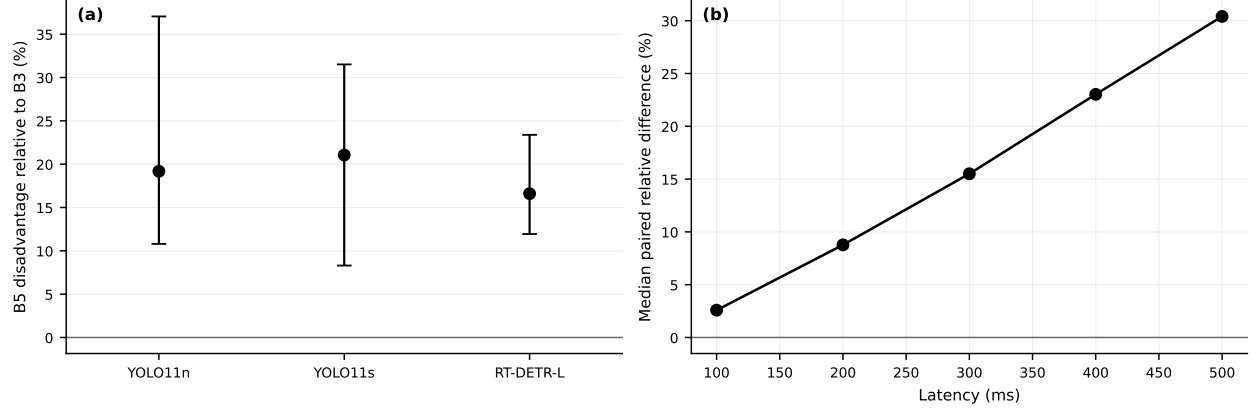


Fig. 1. Architecture and latency robustness. (a) Paired B5 disadvantage relative to B3 at the frozen 300-ms high-yaw endpoint; intervals are 10,000 paired log-cluster bootstrap intervals. (b) Median paired relative difference across all-yaw latency horizons.

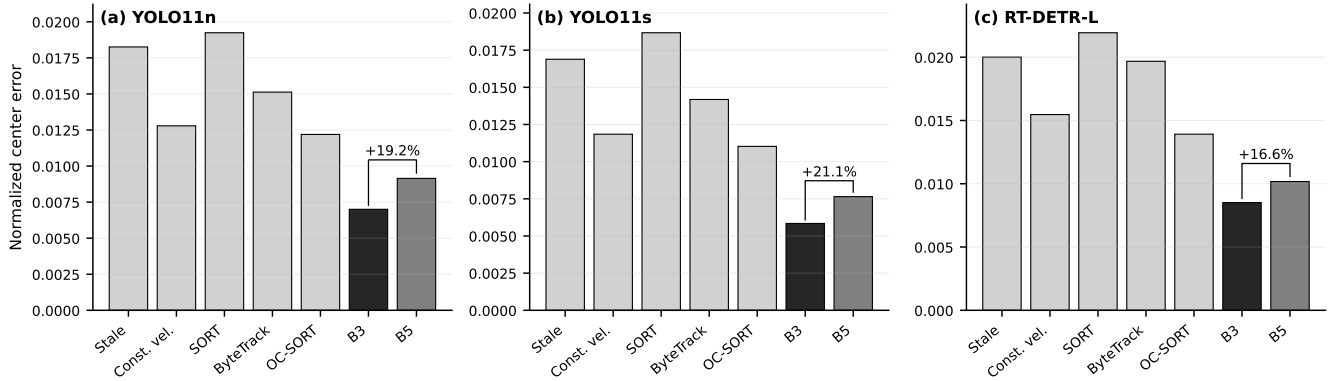


Fig. 2. Causal baseline comparison at 300 ms in high-yaw held-out logs for (a) YOLO11n, (b) YOLO11s, and (c) RT-DETR-L. Bar heights are medians of within-log normalized center error; the two geometric models under comparison (B3, B5) are shaded darker than the causal baselines. The bracket links B3 and B5; its percentage is the paired log-level relative contrast, which differs from the ratio of the displayed aggregate medians.

The second check operates in the image space where error is scored and extends to RT-DETR-L (Table IV). Because B5 adds an injected correction to the B3 box, the displacement that would zero the error is the residual between the B3 box center and the target center, and the sign of the inner product between injected correction and residual states whether the estimate moved the box toward or away from the truth. The correction pointed toward the residual for 57.75% of YOLO11n, 58.03% of YOLO11s, and 56.62% of RT-DETR-L objects. This exceeds chance and matches the oracle evidence that object

motion carries real signal, but the median injected correction of 9.0–13.6 px is of the same order as the 17.5–22.4 px residual it targets, so the 42% of corrections that point away displace the box by roughly the amount they were meant to remove. Weak directional accuracy at full correction magnitude is sufficient to produce net harm.

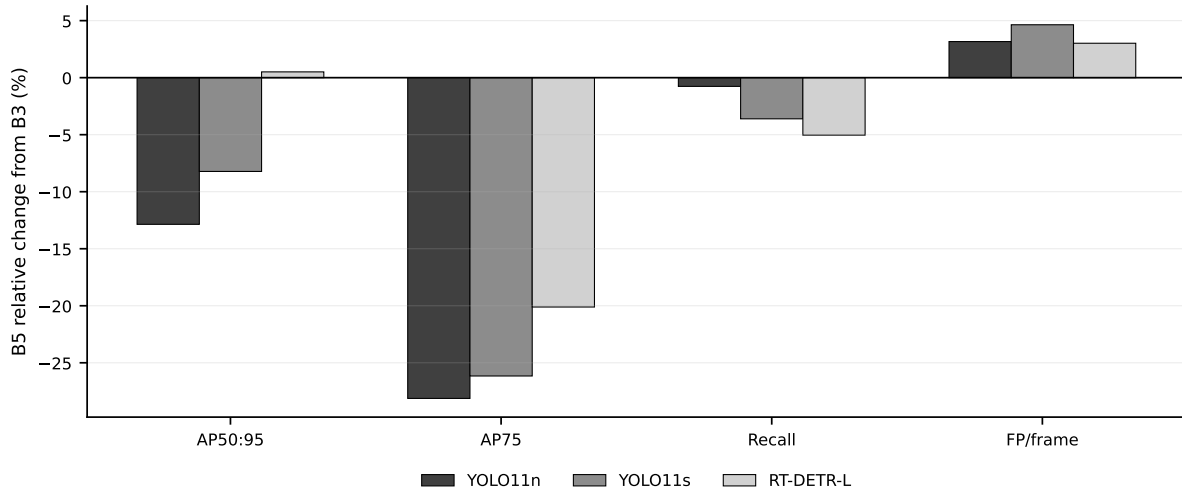


Fig. 3. Pooled full-frame current-time metrics at the frozen endpoint. Negative values for AP and recall favor B3 because B5 is lower; positive false positives per frame also favor B3. Paired log-level $AP_{50:95}$ intervals included zero for all detectors.

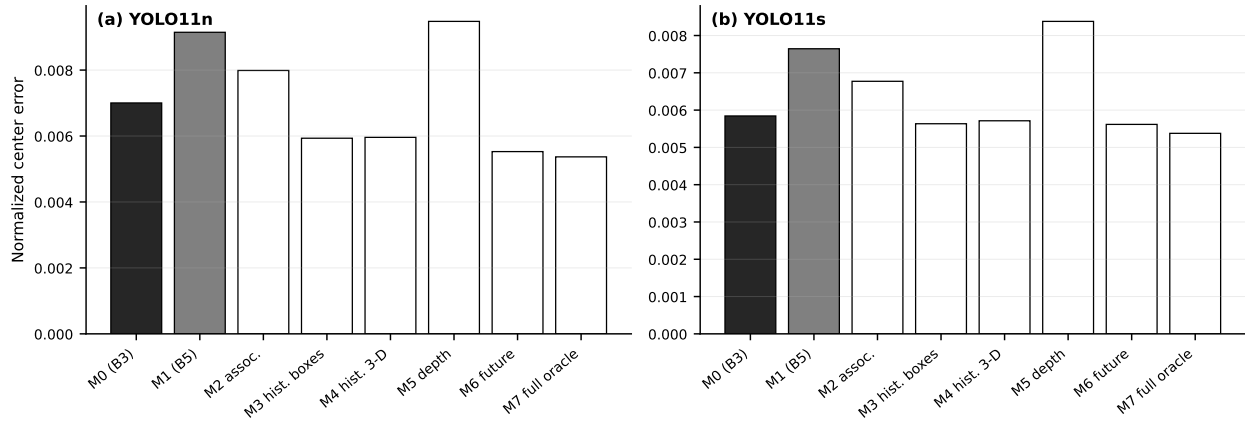


Fig. 4. Component substitutions at 300 ms in high-yaw held-out logs for (a) YOLO11n and (b) YOLO11s. Filled bars are the deployed models (M0 is B3; M1 is B5). Open bars replace one component with oracle information: association (M2), historical boxes (M3), historical 3-D positions (M4), source depth (M5), future displacement (M6), and all components (M7). Future-displacement models are diagnostic and nondeployable.

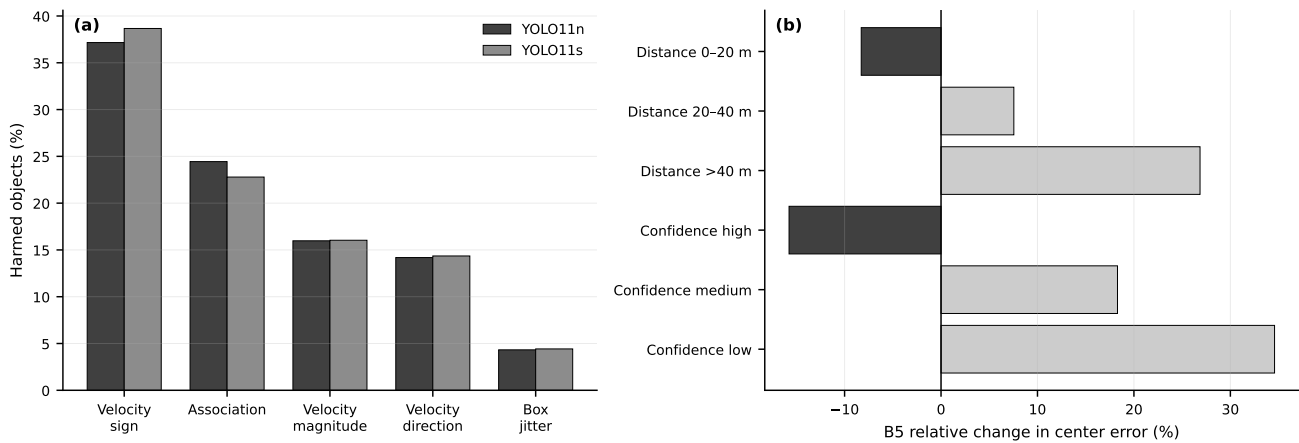


Fig. 5. Diagnostic mechanisms and exploratory operating strata. (a) Fixed hierarchical classifications among objects harmed by B5. (b) Relative B5 center-error change in distance and detector-confidence strata; dark bars indicate strata favoring B5 (negative values), light bars indicate strata favoring B3.

TABLE III
EXPLORATORY CLASS-STRATIFIED RESULTS AT THE FROZEN 300-MS HIGH-YAW ENDPOINT

Detector	Class	Logs	Objects	B3 error	B5 error	Paired B5 disadvantage (95% CI)	Logs favoring B5	Recall B3/B5
YOLO11n	Vehicle	40	10,562	0.006707	0.008453	20.48% (11.72–37.21%)	6/40	0.601/0.561
YOLO11n	Ped./cyclist	23	527	0.010861	0.013090	2.60% (0.00–16.80%)	7/23	0.286/0.299
YOLO11s	Vehicle	40	12,668	0.005802	0.007427	19.01% (10.37–30.08%)	7/40	0.626/0.528
YOLO11s	Ped./cyclist	25	894	0.009622	0.011366	16.63% (0.28–65.80%)	6/25	0.298/0.185
RT-DETR-L	Vehicle	40	21,802	0.007281	0.008738	15.92% (7.68–22.47%)	8/40	0.388/0.374
RT-DETR-L	Ped./cyclist	29	1,872	0.010402	0.013935	26.64% (1.49–44.53%)	8/29	0.095/0.125

Paired disadvantages are medians of within-log relative differences and therefore need not equal ratios of the displayed aggregate medians. Intervals are based on 10,000 paired log-cluster bootstrap replicates. RT-DETR-L rows were reconstructed from the frozen low-level prediction and target tables; the reconstruction reproduces the published YOLO class values to a maximum absolute difference of 1.7×10^{-18} , and per-class comparison counts sum to the totals in Table II.

TABLE IV
IMAGE-SPACE DIRECTION AGREEMENT AT THE FROZEN ENDPOINT

Detector	Eligible	Toward	Away	Corr./resid. (px)
YOLO11n	10,839	57.75%	42.25%	13.6 / 18.1
YOLO11s	13,228	58.03%	41.97%	11.4 / 17.5
RT-DETR-L	22,735	56.62%	43.38%	9.0 / 22.4

Objects whose injected correction or residual fell below one pixel were treated as directionally ambiguous and excluded (5.1%, 6.1%, and 11.6% of objects). The final column reports median injected correction magnitude and median residual magnitude.

TABLE V
PROSPECTIVE REPLICATION ON THE EXTENSION COHORT

Detector	Logs	B3 error	B5 error	Disadvantage (95% CI)
YOLO11n	9	0.006143	0.008478	29.21% (17.80–40.77%)
YOLO11s	33	0.006491	0.007269	20.35% (12.20–24.67%)
RT-DETR-L	15	0.007574	0.008746	10.90% (3.39–31.17%)

G. The reversal replicated prospectively and the frozen gate did not provide a remedy

The extension cohort, held out from every analysis reported above, reproduced the reversal for all three detectors at the frozen endpoint (Table V). Paired disadvantages were 29.21% for YOLO11n over 9 logs, 20.35% for YOLO11s over 33 logs, and 10.90% for RT-DETR-L over 15 logs, with all three bootstrap intervals excluding zero. The RT-DETR-L extension logs supply the prospective architecture check that the gate analysis could not.

The gate’s feature audit found no forbidden evaluation-time inputs; an earlier leakage warning traced to a script that scanned the YAML list of forbidden patterns rather than the evaluated feature list. The frozen selector chose B5 for 0% of eligible extension comparisons, making its output identical to always using B3, and failed its prespecified requirement of improving at least 70% of extension logs. The gate consumes thirteen instrumented source-time features that the frozen pipeline emitted for the YOLO detectors only, so RT-DETR-L extension logs were scorable for the B3-versus-B5 contrast but not for gate selection. The gate result therefore rests on two detector capacities of one family.

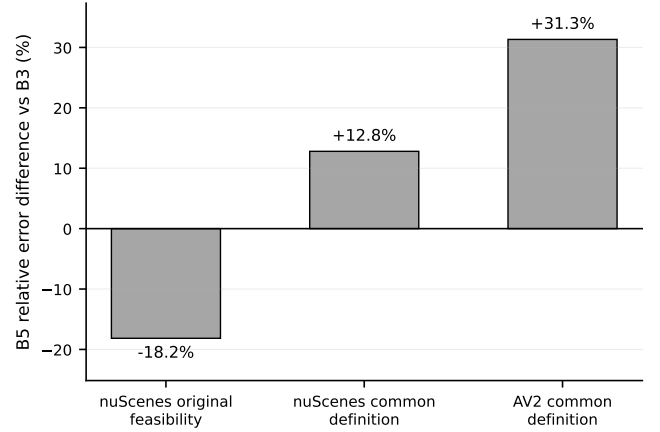


Fig. 6. Cross-dataset harmonization at 500 ms with YOLO11n. Negative values favor B5; positive values favor B3.

H. Cross-dataset harmonization and integrity checks

The initial nuScenes feasibility analysis reported an 18.15% apparent B5 benefit at 500 ms using ground-truth instance-linked history and a weaker ego comparator. Re-evaluation with the common AV2 definitions produced a 12.80% B5 disadvantage in nuScenes and a 31.33% disadvantage in AV2 at the same 500-ms all-yaw comparison (Fig. 6). Comparator strength and association provenance, not intrinsic dataset disagreement, explained the direction change.

Median per-object propagation times were 0.20–0.33 ms for B3/B5 across detectors; OC-SORT prediction took 0.22–0.24 ms. In a separate 2,000-frame benchmark excluding detector inference and input/output, median total post-detection overhead was 1.26 ms per frame for B3 and 1.14 ms for B5, with 95th percentiles of 2.51 and 2.14 ms. Because association, velocity estimation, and gate stages were not timed separately, the small difference between total B3 and B5 times is not interpreted. The core leakage audit found no future-information flags in deployable predictions and no associations after the source timestamp. The provenance audit found no duplicate rows, incomplete T0–T6 sets, invalid coordinates, or detector/path mismatches; 49 headline recalculations agreed to a maximum absolute difference of 5×10^{-14} .

VI. DISCUSSION

The central finding of this study is that a physically motivated component, estimated object motion, consistently degraded a causal latency-compensation system when applied to every detection. The degradation appeared in two capacities of one detector family and in the architecturally distinct RT-DETR-L [17], so it cannot be attributed to the peculiarities of a single network. Nor can it be attributed to a weak reference: at the same frozen endpoint, ego-motion-only propagation also outperformed the causal predictions of SORT [5], ByteTrack [6], and OC-SORT [7], which represent the standard image-space treatment of exactly this extrapolation problem. The expectation inherited from pose-aware propagation, that modeling more of the true motion should reduce error [10], failed in a system where every component was fixed before the outcome was examined.

The component substitutions identify where the expectation breaks down. Let d_o denote the true residual object displacement that remains after ego-motion correction, and let $\hat{d}_o = d_o + \epsilon$ denote its causal estimate. Adding $\hat{d}_o$ reduces error only when the image-space displacement induced by ϵ is smaller than the residual incurred by omitting d_o altogether. Substituting oracle future displacement satisfied this condition and improved on B3 by 11.11% and 9.85% for the two YOLO detectors, which confirms that the residual signal exists and is worth recovering. The deployed estimator violated the same condition: velocity-sign flips and incorrect association together accounted for roughly 60% of harmed objects, meaning the injected correction frequently pointed the wrong way rather than merely overshooting. The direction-agreement analyses reached the same conclusion without relying on the hierarchical labels. After removing near-stationary objects, estimated ground-plane motion occupied the correct half-plane 51.08% and 51.41% of the time, and in image space the injected correction pointed toward the residual it had to remove for 56.62–58.03% of objects while carrying a magnitude comparable to that residual. Because constant-velocity extrapolation scales linearly with the horizon, this injected error also grew faster than the recoverable signal, from a 2.58% penalty at 100 ms to 30.40% at 500 ms. The underlying asymmetry is the one anticipated in Section II: ego motion is measured once per frame from platform poses, whereas object motion must be estimated per object from noisy, intermittently associated detector history.

The full-frame evaluation fixes the boundary of this claim. B3 was superior on the localization-sensitive quantities (matched-object center error, IoU, AP_{75}) and on recall and false-positive burden, but paired $AP_{50:95}$ contrasts included zero for all three detectors, and no propagation-only method can produce an object that first appears during the latency interval. Ego-motion-only propagation is therefore the more reliable transport rule for boxes that already exist; it does not by itself address the broader streaming-perception problem of anticipating future frames [1]–[3].

These results do not imply that object motion should never be modeled. In the exploratory strata, B5 outperformed B3 for objects within 20 m and for high-confidence detections, conditions in which lidar depth support is dense and associ-

ation is comparatively unambiguous, while its disadvantage concentrated beyond 40 m and in the low-confidence stratum. Those observations delineate a plausible operating region, but they do not constitute a selection policy: the frozen causal gate, trained and thresholded before any extension outcome was seen, never selected B5 and therefore delivered no improvement over using B3 everywhere. The open problem is a selector that identifies trustworthy association and velocity estimates prospectively and survives validation on an untouched cohort; until such a selector exists, measured ego motion alone remains the defensible default.

The class analysis extends the claim to vulnerable road users. The B5 disadvantage held for vehicles across all three detectors, so the primary result was not created by pooling heterogeneous road users. The pedestrian and cyclist stratum was initially uninformative under the YOLO detectors, which recovered only 527 and 894 comparisons and produced intervals that touched zero or spanned tens of percentage points. RT-DETR-L detects small objects at higher density and supplied 1872 comparisons across 29 logs, enough to resolve a 26.64% disadvantage with an interval excluding zero. The earlier imprecision was a detector-recall limitation rather than evidence of a class-specific exception, and object-motion augmentation harms localization for pedestrians and cyclists in the same direction as for vehicles.

Finally, the cross-dataset harmonization explains how the opposite conclusion was reached in the first place. The nuScenes feasibility analysis [15] granted the object-motion estimator ground-truth instance-linked history and compared it against a weaker ego-motion baseline, and under those conditions B5 appeared 18.15% better. Re-evaluating both datasets under common definitions, detector-derived history and the calibrated pose-based comparator, reversed the sign within nuScenes itself. Apparent benefit from a pipeline component can therefore originate in privileged information access or in comparator strength rather than in the component, and mechanistic credit should be assigned only after both are matched.

A. Limitations

The primary analysis used one forward camera and AV2 source-time lidar support, and does not evaluate end-to-end 3-D detection, learned future-frame detectors, or complete MOT metrics such as HOTA and identity switches. Registration-based camera-motion-compensation trackers [8], [9] were not run as separate baselines; B3 uses measured poses and lidar depth and therefore approximately upper-bounds what registration-based compensation could deliver on the same detections, but pose-free compensation performance was not quantified. Ground-truth identities constructed evaluation pairs but were unavailable to deployable predictions. Forty of 50 held-out logs entered the high-yaw endpoint; the remainder lacked eligible matched comparisons. The hierarchical mechanism labels and the ground-plane velocity comparison depend on estimator internals that the frozen pipeline recorded for the YOLO detectors only, so those two diagnostics cover two of three architectures; the class-stratified and image-space direction analyses cover all three. Gate selection likewise

requires the instrumented features and was scored on the YOLO extension logs only. Aggregate runtime measurements showed millisecond-scale post-detection overhead but did not isolate association, velocity-estimation, or gate costs. The beneficial near and high-confidence strata are exploratory and require external confirmation.

VII. CONCLUSION

Across YOLO11n, YOLO11s, and RT-DETR-L, universal ego-plus-estimated-object-motion propagation worsened the frozen 300-ms high-yaw localization endpoint relative to ego-motion-only propagation, and B3 outperformed strong causal tracker predictions. Oracle substitutions showed that object motion is useful in principle, but association and velocity-estimation errors exceeded its correction signal; an independent direction analysis found that estimated nonstationary motion pointed into the correct half-plane only about 51% of the time. The reversal held for vehicles under all three detectors and for pedestrians and cyclists once a detector with sufficient small-object recall supplied enough comparisons to resolve the stratum, and it reproduced on a prospective extension cohort. Full-frame results supported the localization conclusion for AP₇₅, recall, and false positives; AP_{50:95} remained mixed. Nearby and high-confidence objects may define a selective operating region, but the frozen causal gate failed to identify it. For this system, measured ego motion is the defensible default; object-motion augmentation should be enabled only by a prospectively validated reliability criterion.

DATA AND CODE AVAILABILITY

AV2 and nuScenes are publicly available under their respective licenses [14], [15]. Analysis code, frozen log lists, configuration files, aggregate tables, figure source data, and audit records are released in a versioned repository and archived with a DOI: v1.0.1, *When More Motion Modeling Hurts: Causal Latency Compensation for Streaming Object Detection*, <https://doi.org/10.5281/zenodo.21446493> (DOI: 10.5281/zenodo.21446493).

AUTHOR CONTRIBUTIONS

Antonio Pereira Jr.: conceptualization, methodology, software, formal analysis, investigation, visualization, writing—original draft, and writing—review and editing.

CONFLICT OF INTEREST

The author declares no competing interests.

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